Housing First or Treatment First?

The model, its parameters, its calibration, and what the referees still want

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Abstract

A complete statement of the rebuilt model: environment, preferences, budget, every transition, the household problem, and the equilibrium. Section 3 lists *every* parameter with its symbol, its role, and either its external source or the moment that pins it. Sections 5 to 8 cover the two-stage calibration, the untargeted moments used for validation, and the mechanism decomposition that replaces the static appendix. Section 9 lists, referee by referee, every comment that is still outstanding.

1 Overview

The model is a life cycle economy in which households choose housing, drug use, treatment, hours and saving under mortality, employment, mental-health and addiction risk, and in which a housing voucher programme is run under one of two rules. The question is whether the voucher should be conditioned on abstinence.

Relative to the submitted paper the model has been rebuilt rather than patched. Homeownership, the housing market, tenure, mortgages and negative assets are gone; the drug price is now an equilibrium object; treatment exists; mental illness exists; preferences no longer vary with addiction status; labour supply is continuous; and every parameter is attached either to a citation or to a moment. Table 8 in Section 9 maps each change to the comments it answers.

Two conventions used throughout. Time is annual. All monetary quantities are in units of average model income, so "0.25" means a quarter of average income.

2 The model

2.1 Demographics and timing

Households enter at model age $j = 1$ (real age 26) and live at most to $j = J = 60$ (real age 85). They retire at $j_R = 41$ (real age 66). Survival from j to $j + 1$ is $\phi_j(h', d, m)$, defined in (13) below: it depends on age, on the housing chosen this period, on substance use disorder and on mental illness. A household that dies leaves its assets behind; the estate accrues to the government (Section 2.14).

Within a period the household observes its state, chooses hours, housing, drug use, treatment and saving, and consumes. Every shock, productivity, employment, addiction, mental health, voucher receipt and survival, is drawn *after* all of that and determines the state the household wakes up in at $j + 1$. Nothing realised this period changes anything the household did this period, which is what makes those variables state variables rather than choices, and it is why the transition kernels in (11) to (14) condition on the CHOSEN h' and v: the choice comes first, the draw second.

2.2 The state

Two features are deliberate.

Housing entered with is not a state. Survival, employment transitions and the recovery hazard all condition on the *chosen* housing h', and renters face no adjustment cost, so the unit a household arrives in is payoff-irrelevant. Carrying it would replicate the whole problem N_h times; removing it leaves every aggregate unchanged and cuts the solve by a factor of 6.6. It would have to be reinstated if moving costs were added.

Three employment states, not four. UI is drawn on *entry* into unemployment with probability π_{UI} and lasts one year. A first-year non-recipient and a household unemployed for two or more years have identical income (zero) and identical continuation values, so those two states

Table 1: The individual state: 12,960 points per age, 777,600 in total

	Symbol	Points	
Age	j	60	real age 26 to 85, retirement at 66
Financial assets	$a \geq 0$	60	cubic spacing, dense near zero; no borrowing
Labour productivity	y	9	Rouwenhorst discretisation of an AR(1) in logs
Employment	e	3	employed; unemployed *with* UI; unemployed without UI
Substance use disorder	d	2	$d = 0$ clean, $d = 1$ SUD
Serious mental illness	m	2	$m = 0$ absent, $m = 1$ present
Housing voucher	η	2	$\eta = 0$ not held, $\eta = 1$ held

merge *exactly*. This answers the AE's A2: UI expiration matters, and it costs one state rather than two.

2.3 Choices

Each period a household chooses

$$\big(n,\ h',\ v,\ \tau,\ a'\big) \in [0,1] \times \{h_1, \ldots, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A},$$

hours n, housing h' (where h_1 denotes homelessness and $h_2 < h_3 < h_4$ are rental sizes), drug consumption v, treatment take up τ (available only if $d = 1$), and next-period assets a' on the asset grid.

2.4 Preferences

Period utility is

$$u = \frac{\big[c^{\alpha}\,(h')^{\beta}\,(v + b_{dmh'})^{\gamma}\big]^{1-\sigma}}{1-\sigma} \;+\; \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u} \;-\; \kappa_x\big(H + \omega_x H_D\big), \qquad (1)$$

with the Cobb–Douglas weights

$$\alpha = (1-\theta_h)(1-\theta_v), \qquad \beta = \theta_h(1-\theta_v), \qquad \gamma = \theta_v, \qquad \alpha+\beta+\gamma = 1.$$

The weights do not depend on addiction status. Referee 2 (C5), Referee 3 (M2a) and the AE (A8) all objected that in the submitted paper the taste weight on drugs rose with d, so reducing SUD mechanically raised the utility derived from c and h and made addiction-reduction and housing expansion complements "even absent incentive effects". Fixing α, β, γ removes that channel completely. It is also a *commitment*: with preferences invariant, becoming addicted no longer relaxes any constraint, so every private cost of SUD must now come from mortality, employment and the budget. Section 8 tests exactly that.

Only the Stone–Geary shifter varies. The AE's suggested alternative (A8) was to vary b instead, and that is what the model does:

$$b_{dmh'} = b_d \cdot \rho_m^{\mathbf{1}\{m=1\}} \cdot \rho_H^{\mathbf{1}\{h'=h_1\}}, \qquad d \in \{0,1\}, \quad \rho_m, \rho_H \in (0,1]. \tag{2}$$

A lower b raises the marginal utility of the first units of use. Three separate things are going on here and each is doing a job.

1. b_0 *versus* b_1. A single b cannot match both the participation rate and the drug-expenditure share. A b large enough to deter clean households pushes the unconstrained interior choice v^* below zero, which drives expenditure by households *with* SUD to the corner. Two values separate the moments: b_0 pins participation, b_1 keeps the choice interior so that θ_v pins the expenditure share.

2. $\rho_m < 1$*: self-medication.* Poor mental health raises the marginal value of the first units of use. This is the channel Referee 1 asks about in comment 1(a).

3. $\rho_H < 1$*: the housing-access channel.* This is the only parameter in the model that governs the elasticity of drug use with respect to housing access, the object Referee 2 (C1) calls the crux on which all the welfare results rest, and which Referee 3 and the AE (A13) name independently. Without it that moment is not merely untargeted, it is *untargetable*: housing a household only raises its resources, drugs are a normal good, and the model would predict that housing *raises* use, the opposite sign to Milby et al. (2005). Section 5.2 calibrates ρ_H to that trial.

Leisure. ω_ℓ is the weight on leisure and σ_ℓ its curvature, which pins the Frisch elasticity at $1/\sigma_\ell$.

ℵ, the flow value of being alive. ℵ is what makes mortality risk deter drug use: without it, dying is not costly and the model's central mechanism disappears. It also has to satisfy a condition that is easy to overlook, see Section 7.

$\bar{\tau}^u$, the utility cost of treatment. Treatment take up costs utility, not resources. A resource cost comes out of the same budget that buys food and rent, so the households with the strongest case for treatment, the poorest, would be exactly the ones unable to afford it, which is not how publicly funded treatment works. $\bar{\tau}^u$ is the stigma, effort and time of entering a programme whose monetary cost somebody else bears.

The externality. H is the aggregate homelessness rate and H_D the share of the population both homeless and with SUD. Referee 2's wording is specific, "a large homeless population, *especially if many are addicted to drugs*, creates a negative externality beyond the fiscal burden", so the disamenity is weighted by composition through ω_x rather than depending on the headcount alone. Households take H and H_D as given, which is what makes this a spillover rather than a private cost, but both are equilibrium objects, so policy moves them and the movement feeds back through the value of life into drug decisions. Section 8.6 calibrates κ_x from a contingent-valuation study rather than leaving it at zero.

2.5 Income, taxes and transfers

Gross income depends on age, employment, productivity and health:

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\, y & j \geq j_R \quad \text{(social security)} \\ n\, y\, \varepsilon_j\, \rho_d^{\mathbf{1}\{d=1\}} \rho_m^{\mathbf{1}\{m=1\}} & j < j_R,\ e=1 \quad \text{(employed)} \\ \varrho_{UI}\, y\, \varepsilon_j & j < j_R,\ e=2 \quad \text{(UI, first year only)} \\ \zeta_{DI}\, \bar{y}_{DI}\, \bar{Y} & j < j_R,\ e=3,\ m=1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{3}$$

with, in order of appearance: y the persistent productivity state; ε_j the deterministic age profile of labour efficiency; $\rho_d \leq 1$ and $\rho_m \leq 1$ the productivity penalties from substance use disorder and serious mental illness, applied *conditional on working*; ϱ_{UI} the unemployment-insurance replacement rate, paid on the household's own productivity and only in the first year of a spell; ζ_{DI} the share of the non-employed with serious mental illness who receive disability insurance; $\bar{y}_{DI}$ the disability benefit expressed as a share of average income; and $\bar{Y}$ average income in the economy. Retirement income is a flat 40% replacement of the productivity state, with no health penalty, social security is not means tested on health.

Why the penalties are conditional on working. This is what allows an *employed* household to have income low enough to choose homelessness. An unemployment effect alone cannot do that, and Referee 1's comment 2(c)(i) is precisely that the model produced a homeless population that was 100% out of work against 40 to 52% in the data.

Labour income is taxed with the Heathcote, Storesletten, Violante function and capital income at a flat rate:

$$y^n = \tau_\ell\, (y^g)^{\xi} \ + \ (1-\tau_k)\, r\, a. \tag{4}$$

Two transfers. A *food-stamp floor* tops means tested income up to $\underline{c}\,\bar{Y}$:

$$F = \max\Big\{0,\ \underline{c}\,\bar{Y} - \big(y^g + \max\{0, ra\}\big)\Big\}, \tag{5}$$

and a *housing voucher*, paid to voucher holders and only while renting, covering rent above a tenant contribution of κ_r of gross earnings:

$$Vch = \mathbf{1}\{\eta = 1\}\mathbf{1}\{h' \neq h_1\} \max\big\{0,\ p_h h' - \kappa_r\, y^g\big\}. \tag{6}$$

A homeless voucher holder does not lose the voucher. Retention depends on income eligibility, not on housing, so a household that gives up its unit keeps its entitlement and the voucher simply pays nothing until it rents again. With homeownership gone there is no other tenure for it to be spent on, so this is both the realistic rule and the simple one.

Both are means tested on *gross* income, as in the programmes themselves. Both are also clawed back, at a rate of one for food stamps and κ_r for the voucher, and that is what generates the welfare trap discussed under labour supply below.

2.6 The budget constraint

$$c + p_v v + a' \;=\; \underbrace{y^n + a}_{\text{after-tax income and assets}} \;-\; \underbrace{p_h h' \mathbf{1}\{h' \neq h_1\}}_{\text{rent}} + Vch + F, \qquad c > 0,\;\; v \geq 0,\;\; a' \geq 0. \tag{7}$$

Note what is *not* in (7): treatment, which costs utility; and bequests, which go to the government.

The closed form split of c and v. Given a' and h', let R denote the resources remaining. Cobb–Douglas preferences over $(c,\, v + b)$ give

$$c = \frac{\alpha}{\alpha + \gamma}\big(R + p_v b\big), \qquad v = \max\Big\{0,\; \frac{\gamma}{(\alpha + \gamma)p_v}\big(R + p_v b\big) - b\Big\}, \tag{8}$$

so the inner allocation never needs a numerical search. The max is the corner at which the household abstains because the Stone–Geary endowment already exceeds its desired intake, which is exactly how a large b deters use.

2.7 Labour supply

Hours are continuous and satisfy the intratemporal first-order condition

$$u_c(c)\,\frac{\partial R}{\partial n} \;=\; \omega_\ell (1 - n)^{-\sigma_\ell}. \tag{9}$$

Two properties of (9) matter substantively.

The residual is not monotone: the welfare trap. From (5) and (6), $\partial R/\partial n$ jumps *up* at the hours where each transfer is exhausted, because the clawback stops. Below the food-stamp exhaustion point the marginal return to an hour of work is close to zero. So "work a little and keep the transfer" and "work properly" are genuinely distinct local optima, and the household compares them. The solution algorithm splits $[0, 1]$ at the kinks and solves each piece exactly; bracketing once would silently pick whichever optimum happened to be bracketed.

Eligibility is a step in the continuation value. Voucher eligibility is lost discretely when means tested income crosses κ_V times the median. That does not kink the budget, so (9) is continuous across it, but it shifts the continuation value, so the piecewise optimum is clamped to each side of the threshold and both are evaluated.

Referee 3 (m1) wanted labour supply kept and its role made visible; the AE (A3) questioned whether it was needed. The decisive reason to keep it is not the welfare cost of tax financing but the programme itself: **voucher eligibility and the voucher payment are both functions of earnings**, so hours are the margin through which a household controls whether it qualifies and how much it receives. Without an hours choice the voucher's means test would be something that happens to a household rather than something it responds to, and the labour supply distortion created by a 30% tenant contribution on top of the food-stamp clawback, the welfare trap above, would be invisible. That distortion is a first-order part of what a housing voucher does. Making hours continuous and exact resolves the tension in Referee 3's favour, at a cost of a factor of three in solve time.

2.8 Housing and homelessness

Housing is a four-point grid. h_1 is homelessness, with a small service flow h_{HL} and no rent. The concept is **literal homelessness**, sleeping rough or in an emergency shelter, the HUD point in time definition. It *excludes* doubling-up and couch-surfing, which are far more common and would put the rate an order of magnitude higher. The AE's A16 notes that the share of the homeless with SUD depends heavily on which definition is used, and it does: the 21.7% target is measured on the literal-homelessness population, so the model must target the same population. The sensitivity to a broader definition is reported. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{10}$$

so two parameters describe the whole grid: the level $\underline{h}$ and the spread ς.

Rent is normalised. Housing services h' enter utility (1); the rent bill $p_h h'$ enters the budget (7). The two are nearly collinear: rescaling the grid by λ and the price by $1/\lambda$ leaves every budget constraint untouched and changes utility only by the factor λ^{β} inside the consumption bundle, which affects behaviour solely through the curvature interaction with the additive leisure and $\aleph$ terms. Identification of the level therefore runs through a weak channel and should not be relied on. Setting $p_h = 1$ makes the *grid* the calibrated object, and $\underline{h}$, ς and h_{HL} are then pinned by three moments of the housing-burden distribution and the homelessness rate. This is what Referee 2's comment D1 asked for: in the submitted paper the grid had seven points, four of them uncalibrated and listed in a footnote.

2.9 Drug use, addiction, recovery and treatment

A household either abstains ($v = 0$) or uses ($v > 0$, chosen continuously according to (8)). Transitions in d:

$$\Pr(d' = 1 \mid d = 0) = \pi_d \cdot \mathbf{1}\{v > 0\}, \qquad \Pr(d' = 0 \mid d = 1) = \pi_r(h', \tau, m). \tag{11}$$

A single addiction hazard. A clean household that uses becomes addicted with probability π_d *regardless of how much it consumes.* There is no light/heavy threshold $\bar{v}$ and no consumption-dependent hazard; the 6.48 heavy-to-light ratio is gone. This removes two of the parameters Referee 3 (M2b) asked to have justified, at the cost of one channel: intensity no longer affects the risk of addiction, only the budget.

Recovery and treatment. The recovery hazard is

$$\pi_r(h', \tau, m) = \min\Big\{1,\ \bar{\pi}_r \cdot \mu_H^{\mathbf{1}\{h' = h_1\}} \cdot \mu_m^{\mathbf{1}\{m=1\}} \cdot g^{\tau}\Big\}, \tag{12}$$

with $\mu_H < 1$ (recovery is harder when homeless), $\mu_m < 1$ (harder with mental illness) and $g > 1$ the treatment effect. $\tau \in \{0, 1\}$ is a choice, available only to households with $d = 1$, at the utility cost $\bar{\tau}^u$ in (1).

Referee 1's comment 1(b) and the AE's A10 are the same point and it is the most damning gap in the submitted paper: the policy is called Treatment First and there was no treatment in the model. There is now, it is a decision rather than a mechanical hazard, and g, through which the entire Treatment First result runs, is disciplined in Section 5.2.

The maximum experimentation age. In the baseline a clean household may initiate only up to $j \leq j_E$. Referee 1's smaller comment 1 notes this is in tension with the U-shaped age profile of use in Table 4(a). It is now a variable rather than a hard-coded parameter, so the model can be run without it; Section 7 reports the age profile both ways.

2.10 The private costs of SUD: mortality, employment, and the homelessness loop

This subsection is the heart of the mechanism and it deserves to be stated explicitly rather than buried in a parameter table.

Mortality. Survival is

$$\phi_j(h', d, m) = 1 - \min\Big\{1,\ \big(1 - \bar{\phi}_j\big) \cdot \kappa_{\phi H}^{\mathbf{1}\{h'=h_1\}} \cdot \kappa_{\phi d}^{\mathbf{1}\{d=1\}} \cdot \kappa_{\phi m}^{\mathbf{1}\{m=1\}}\Big\}, \tag{13}$$

where $\bar{\phi}_j$ is the baseline age profile of survival. SUD multiplies the death hazard by $\kappa_{\phi d}$, homelessness by $\kappa_{\phi H}$ and mental illness by $\kappa_{\phi m}$. Because $\aleph > 0$, a higher death hazard is a real cost, and it is larger for households with more remaining life and more to consume, which is the sorting the paper's mechanism relies on.

Employment. Job finding and separation are

$$\pi^{ue} = \min\big\{1,\ \bar{\pi}^{ue}\, \kappa_{ueH}^{\mathbf{1}\{h'=h_1\}} \kappa_{ued}^{\mathbf{1}\{d=1\}} \kappa_{uem}^{\mathbf{1}\{m=1\}}\big\}, \qquad \pi^{eu} = \min\big\{1,\ \bar{\pi}^{eu}\, \kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}} \kappa_{ued}^{-\mathbf{1}\{d=1\}} \kappa_{uem}^{-\mathbf{1}\{m=1\}}\big\}, \tag{14}$$

so each multiplier below one simultaneously makes work harder to find and easier to lose. On separation the household draws UI with probability π_{UI}:

$$e = 1 \ \longrightarrow\ \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu}\pi_{UI} \\ e' = 3 & \text{w.p. } \pi^{eu}(1 - \pi_{UI}) \end{cases} \qquad e \in \{2, 3\} \ \longrightarrow\ \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}. \end{cases}$$

The homelessness loop. Homelessness enters (13), (14), (12) and (2): it raises mortality, worsens employment, lowers the recovery hazard, and raises the marginal utility of drug use. Those four together are what make homelessness and SUD mutually reinforcing, and shutting them off jointly is arm 4 of the mechanism ladder.

What SUD does *not* do. In the baseline there is no direct wage penalty conditional on working, $\rho_d = 1$. The AE (A20) asks about direct productivity effects. The robust findings are on the employment margin, roughly 50% more unscheduled absence and 44% higher turnover, captured by κ_{ued}, while the wage penalty conditional on working is contested, with parts of the older literature finding positive effects for casual use. The baseline therefore imposes none and says so; a version calibrated to the SUD income ratio is reported as an alternative.

2.11 Serious mental illness

Why the mental health transition does not depend on housing, and why that is now a result. The model lets homelessness lower SUD recovery (λ_{HL}) but leaves the mental health chain independent of housing. That asymmetry looks like an omission and it is not. In the At Home/Chez Soi randomised trial, $N = 1198$ homeless adults with mental illness across four Canadian cities, assignment to scattered site housing with intensive case management produced **no detectable effect on mental illness symptom severity at any wave** — -0.96 ($p = .16$), -0.51 ($p = .46$), 0.18 ($p = .79$), 0.57 ($p = .44$) — and none on the Recovery Assessment Scale, 0.09 ($p = .92$). What the same trial *does* find, large and significant, is housing stability, its primary outcome, and condition specific quality of life, 5.91 (3.41 to 8.41, $p < .001$).

Three things about how much weight this bears. The design is right for the question: it randomises housing access among exactly the population the parameter would govern. The estimate is if anything *diluted*, because the control arm also achieves some housing, which makes a null more rather than less telling. And the outcomes are symptom scales over 24 months rather than diagnostic remission over a lifetime, so the honest claim is that housing does not measurably speed recovery from serious mental illness on that horizon, not that it could never matter. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

The same trial appears to contradict Milby, and the resolution is this paper's thesis. At Home/Chez Soi also reports severity of substance use problems at 1.04 (0.89 to 1.21, $p = .66$): unconditional housing does not reduce substance use either. Yet b_{HL} is calibrated to Milby, where housing *did* raise abstinence. The two are reconciled by what distinguishes them: Milby's housing was **contingent on abstinence**, At Home/Chez Soi's is **unconditional**. Contingency is doing the work, not housing per se. Two trials with opposite designs therefore identify exactly the margin this paper is about, and jointly they are far better evidence for the mechanism than either is alone.

m is an exogenous two-state Markov chain. It represents *serious* mental illness, not any mental illness: AMI covers roughly 23% of US adults and includes mild anxiety and depression, whereas SMI is about 5.5% and is defined by substantial functional impairment. A state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is *derived* from the target prevalence p_m and the recovery rate λ_m,

$$\text{onset} = \lambda_m \frac{p_m}{1 - p_m}, \tag{15}$$

so calibrating persistence never disturbs prevalence.

SMI affects six objects: productivity conditional on working (ρ_m), job finding and separation (κ_{uem}), mortality ($\kappa_{\phi m}$), the recovery hazard (μ_m), the Stone–Geary shifter (ρ_m in (2)), and eligibility for disability income (ζ_{DI} in (3)).

And causation runs the other way too: SUD raises the hazard of SMI. The onset rate is multiplied by $\iota > 1$ for a household with $d = 1$:

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m \cdot \iota^{\mathbf{1}\{d=1\}}. \tag{16}$$

The evidence is strong. Substance-induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near 25% (Murrie et al., *Schizophrenia Bulletin*), and in the Danish registers

any substance use disorder predicts conversion to schizophrenia with a hazard ratio of 1.34, amphetamine use disorder 1.90 and opioid use disorder 2.74. The central value is $\iota = 2$ with a range of 1.3 to 2.7.

This channel matters twice over, and the submitted paper had neither half of it. *Privately* it is a further cost of addiction, so it belongs in the mechanism ladder alongside mortality and employment. *Fiscally* it is a further externality, and a compounding one: SUD raises SMI, SMI raises disability receipt, lowers employment, raises mortality and raises homelessness, so the chain SUD $\rightarrow$ SMI $\rightarrow$ DI and lost tax runs entirely outside what the household internalises beyond its own utility. It is measured in §8.6.

Because onset is no longer common to everyone, it can no longer be derived from prevalence: the base rate is calibrated so that population prevalence still lands on 5.5% while the multiplier operates.

The specification follows İmrohoroğlu and Zhao (2022) structurally, not just numerically: in their model labour-market shocks generate *short* homeless spells while health shocks generate *chronic* homelessness, and expanded means tested disability income is the effective policy for the chronic group. Homelessness driven purely by income risk is transitory; the persistent tail needs a persistent state. This is Referee 1's comment 1(a), and the persistence is also what generates the older homeless population of comment 2(c)(ii).

2.12 The voucher programme and the two policies

A household without a voucher draws one with probability π_V if it is income eligible; a household with one keeps it while eligible. Eligibility is

$$y^g + \max\{0, ra\} \;<\; \kappa_V \cdot \text{median income}. \tag{17}$$

Under **Housing First** that is the whole rule. Under **Treatment First** a voucher is additionally lost with probability p_{det} if the household used drugs, and with probability p_{fp} even if it did not:

$$\Pr(\eta' = 1) = \Pr(\eta' = 1)^{HF} \times \begin{cases} 1 - p_{\text{det}} & \text{if } v > 0 \\ 1 - p_{\text{fp}} & \text{if } v = 0. \end{cases} \tag{18}$$

Referee 1 (4a, 4b) and the AE (A11) both note that perfect observation and enforcement are not credible. p_{det} and p_{fp} are experiment parameters reported as sensitivity, not calibrated, because that is the point.

2.13 The household problem

Let $s = (a, y, e, d, m, \eta)$. The value function solves

$$V_j(s) = \max_{\{n,\, h',\, v,\, \tau,\, a'\}} \Bigg\{ \frac{\left[c^{\alpha}(h')^{\beta}(v + b_{dmh'})^{\gamma}\right]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x\big(H + \omega_x H_D\big) + \beta\,\phi_j(h', d, m)\,\mathbb{E}\Big[V_{j+1}\big(a', y', e', d', m', \eta'\big) \;\Big|\; s, h', v, \tau\Big] \Bigg\} \tag{19}$$

subject to (3) to (7), with $V_{J+1} \equiv 0$: there is no bequest motive, and death is worth zero. The expectation is taken over the independent transitions in y (Rouwenhorst), e (14), d (11), m, and η.

The problem is solved by exact backward induction. Within a drug regime and a housing choice the c and v split is closed form (8) and hours come from (9), so the only search is over a', done on the full grid because the objective is *not* single-peaked in a': voucher eligibility is a step function of income (which includes ra), the food-stamp floor kinks, and housing is discrete.

2.14 Government

The government pays social security, UI, disability, food stamps and vouchers, and bears a fiscal cost χ_H per homeless household and χ_D per household with SUD. It finances this with the labour income tax (4), the capital income tax, and the estates of the deceased. The level parameter τ_ℓ clears the budget:

$$\underbrace{\sum y^g - \tau_\ell \sum (y^g)^\xi + \tau_k \sum \max\{0, ra\} + \text{estates}}_{\text{revenue}} = \underbrace{SS + UI + DI + F + Vch + \chi_H H + \chi_D D + G}_{\text{outlay}}. \tag{20}$$

Estates go to the government. Redistributing accidental bequests as a uniform lump sum is a large unconditional transfer to the poor with welfare consequences of its own, which would contaminate every policy comparison in the paper. Sending them to the budget also removes an equilibrium fixed point. The lump-sum version remains available for a robustness run. This is a change from the submitted paper, where transfers were additionally type-dependent, the AE's A7.

2.15 Drug supply and the price

Drug production has cost $C(V) = \frac{\psi_1}{\psi_2} V^{\psi_2}$, so supply satisfies $p_v = \psi_1 V^{\psi_2 - 1}$ and the price elasticity of supply is $1/(\psi_2 - 1)$. The shifter ψ_1 is normalised so that $p_v = 1$ at the benchmark quantity.

This answers Referee 1 (1c), Referee 3 (M2d) and the AE (A5). Referee 3 worried that endogenising the price "requires suppressing this normalization". It does not: the normalisation moves to ψ_1, the benchmark price is still one, and only counterfactuals move it.

What the evidence says about ψ_2, and why it favours the submitted paper. The maintained assumption in the drug-economics literature is that US supply of cocaine and heroin is *very* elastic. Rhodes et al. (2002), discussed in the National Academies' *Understanding the Demand for Illegal Drugs*, assume it outright on the grounds that these are agricultural products with simple, inexpensive processing, so producers can meet any demand at roughly the current price; the reduced-form demand estimates that literature rests on require it.

There is also a natural experiment. In August 2010 Purdue replaced OxyContin with an abuse-deterrent formulation, pushing a very large mass of demand into heroin, enough to quadruple heroin death rates (Alpert, Powell and Pacula, NBER w24475). Purity-adjusted heroin prices did not rise. They fell slightly between 2010 and 2012. A demand shift of that magnitude leaving price flat is direct evidence of a near-horizontal supply curve.

So the baseline here is a *flat* supply curve and finite elasticities are the robustness direction, not the other way round. That reframes the answer to Referee 1 (1c) and Referee 3 (M2d): the submitted paper's fixed price was the field's standard assumption rather than an unjustified normalisation, and the machinery now exists to demonstrate that relaxing it does not matter.

It is worth recording the companion fact, because it disciplines the second stage. The *demand* elasticity is well estimated: participation elasticities of -0.36 to -0.55 for cocaine and -0.80 to -0.90 for heroin (Saffer and Chaloupka), with conditional heroin demand near -0.80 (Olmstead et al. 2015). The paper's -0.63 target sits inside that range, and the -0.36 to -0.9 robustness band Referee 1 asks for *is* that range.

2.16 Stationary equilibrium

A stationary equilibrium is a value function, policy functions, a distribution Φ, and $(\tau_\ell, p_v, \bar{Y}, \text{median}, H, H_D)$ such that households solve (19); Φ is the invariant distribution induced by the policies and the transitions; the government budget (20) holds; the drug market clears at p_v; and $\bar{Y}$, the median, H and H_D are the aggregates implied by Φ.

Note that H and H_D enter (1), so they are genuine fixed points whenever $\kappa_x > 0$, and that the median enters voucher eligibility, so it is a fixed point too, in a skewed income distribution the median sits well below the mean, and using the mean would materially overstate the eligible population.

3 Parameters

Every parameter in the model. **Status** is one of: **N** normalisation or definition; **L** standard in the literature, cited; **D** set directly from data; **C** internally calibrated, with the moment it is assigned to; **E** determined in equilibrium; **S** reported as a sensitivity rather than calibrated, because no credible point estimate exists.

Table 2: Preferences and prices

Symbol	Meaning	St.	Value, source or target
σ	risk aversion	L	2.0. Attanasio and Weber (2010) survey; the standard life cycle value
σ_ℓ	curvature on leisure	L	3.0; Frisch elasticity 0.5, Chetty (2012)
ω_ℓ	weight on leisure	C	$\rightarrow$ average hours worked, 40% of time. Guvenen et al. (2023)
θ_h	Cobb–Douglas weight on housing	C	$\rightarrow$ aggregate housing cost / income, 21.2%. SCF 2022, ages 26 to 85, all households, owner housing at imputed rent; `data/scf_housing_burden.py`

Symbol	Meaning	St.	Value, source or target
θ_v	Cobb–Douglas weight on drugs	C	→ drug spending / income given SUD, 44%. Carried over from the submitted paper; its construction must be stated explicitly in the appendix (R1 2e)
b_0	Stone–Geary shifter, clean	C	→ drug participation, 5.13%. NSDUH 2024 microdata, ages 26+; `data/nsduh_gradients.py`
b_1	Stone–Geary shifter, SUD	C	→ abstinence rate among homeless adults with SUD, 18.7%. Milby et al. (2005), NO-HOUSING arm: 4.48 consecutive weeks abstinent out of 24. b_1 governs how costly abstention is once a household already has SUD, so one arm of the trial pins the level and the contrast between arms pins the treatment effect. It previously had no target at all, which also left the housing-access ratio undefined, with abstention impossible, the NH arm abstains 0% and the ratio is 0/0
ρ_m	b multiplier, mental illness	C	→ SUD rate given SMI, 9.75%. NSDUH 2023 detailed tables, ages 26+, by Bayes on Tables 6.13B, 6.4B and 5.10B
ρ_H	b multiplier, homelessness	C^2	→ housed/unhoused abstinence ratio, 1.473. Milby et al. (2005), *AJPH* 95(7):1259 to 65, NACH versus NH arms
$\aleph$	flow value of being alive	C	→ **model implied VSL, $7.5m** (Viscusi and Aldy 2003; EPA 2010). It used to target the SUD income ratio, which pins $\aleph$'s effect on WHO uses but not its LEVEL, and at the resulting level 64.6% of the population had $V < 0$, i.e. preferred death, which turns the mortality cost of drug use into a benefit. The SUD income ratio moved to ρ_d

Symbol	Meaning	St.	Value, source or target
$\bar{\tau}^u$	utility cost of treatment	C	$\rightarrow$ treatment take up given SUD, $\approx$24%. NSDUH 2023: about a quarter of people with a past-year SUD received any substance use treatment. **Verify against the detailed tables**
β	discount factor	C	$\rightarrow$ wealth / GDP = 3. BEA fixed assets against GDP; the standard US private-capital-to-output ratio
p_v	drug price	E	clears the drug market; = 1 at the benchmark by the normalisation of ψ_1
r	risk-free rate	L	0.03, Kaplan, Mitman and Violante (2020)
κ_x	externality weight	D	Inverted from Loubière et al.'s eight-country contingent valuation, €57 per adult per year to end homelessness, normalised per unit of homelessness by the European reference rate. §8.6
ω_x	extra weight on homeless-with-SUD	S	**1.0 in the baseline**: the disamenity depends on the headcount alone. No direct estimate exists, R2's wording ("especially if many are addicted to drugs") implies a positive weight and the hedonic split between supportive housing and emergency shelter implies composition matters, but neither pins a number, so the composition-weighted version is an **appendix robustness**, swept over 1 to 4

Table 3: Housing, taxes and transfers

Symbol	Meaning	St.	Value, source or target
h_{HL}	service flow of homelessness	C	$\rightarrow$ homelessness rate, 0.399%. HUD PIT 2023, converted to **households**: 467,020 homeless individuals, one household each, plus 57,563 homeless family households, against 131,332,360 occupied units. The model's agent is a household, so the person rate of 0.195% is the wrong unit; the factor of two is because homeless households are overwhelmingly single person while US households average about 2.5 people. Literal homelessness, excluding doubling up (AE A16)
$\underline{h}$	smallest rental size	C	$\rightarrow$ share with housing cost $> 50\%$ of income, 17.49%. SCF 2022; `data/scf_housing_burden.py`
ς	geometric spread of rental sizes	C	$\rightarrow$ share with housing cost $> 30\%$ of income, 40.62% (SCF 2022). **Data now in hand** for the better target: ACS B25042 (*Tenure by Bedrooms*, occupied units, 2023, $N = 131{,}332{,}360$) gives 13.21% / 24.12% / 62.67% across $\{0, 1\}$, $\{2\}$, $\{3+\}$ bedrooms. All households, not renters: the model has no homeownership, housing is a pure flow at $p_r h$, so an owners analogue is imputed rent and the SIZE distribution over all occupied units is the right object. That choice is worth 2.4$\times$ on the small bin, renters alone give 32.35%, and the bin cut matters more still, $\{3+\}$ at 62.67% against $\{4+\}$ at 23.63%. In `data/us_housing_dist.csv`; not yet wired in as a moment
ξ	tax progressivity (HSV)	L	0.85, Heathcote, Storesletten and Violante (2017)
τ_ℓ	tax level	E	clears the government budget (20) given the distribution

Symbol	Meaning	St.	Value, source or target
τ_k	capital income tax	L	0.15. US long run capital-gains and qualified-dividend rate for the middle brackets
$\underline{c}$	food-stamp consumption floor	C	$\rightarrow$ food stamps / labour income, 0.42%. USDA Food and Nutrition Service, SNAP outlays FY2023, against GDP
κ_V	voucher eligibility threshold	D	0.50 of median income, HUD rule
κ_r	tenant contribution rate	D	0.30, HUD rule
π_V	voucher lottery probability	C	$\rightarrow$ voucher recipients / income-eligible, 26.6%. HUD Picture of Subsidized Households against the income-eligible renter population
p_{test}	probability a voucher holder is tested, TF	S	1.0 at the benchmark, swept downward. Detection now requires a test, so effective detection is $p_{\text{test}}p_{\text{det}}$ and the old behaviour is the $p_{\text{test}} = 1$ corner
c_{test}	cost per voucher holder tested	D	0.0023 of average income. A urine panel runs about \$40 and quarterly testing is roughly \$160 a year against average earnings near \$70,000. It enters the GOVERNMENT BUDGET, where it competes with everything else the government pays for, R1's comment 4(c) is that the \$1 to 2bn estimate is too large to leave in a dismissive footnote
p_{det}	detection probability given a test	S	1.0 in the benchmark, swept downward. No estimate exists and none should: R1 4(a) and AE A11 object precisely to assuming perfect enforcement, so the point is the sweep
p_{fp}	false-positive rate under TF	S	0.0 in the benchmark, swept upward. As above, R1 4(b)

Symbol	Meaning	St.	Value, source or target
χ_H	fiscal cost per homeless household	D/S	**0.4286.** The homeless state h_1 is *literal* homelessness, shelters and the street, not doubling up, so the matching figure in Abramson and Van Nieuwerburgh (2024, NBER w32582, p. 30) is their $2,500 per household per month for the literally homeless, not the $1,775 they blend across the literally homeless and the doubled-up. $30,000 a year against the models $70,000 average income. Components, their footnote 11: shelters, social services, extra policing, hospital emergency care; they call the blended figure *conservative*. **Note the direction**: R2 and R3 both judged the previous 0.252 too HIGH, and the best structural number in the literature is 70% higher again. It is an assumption inside their calibration rather than an independent estimate, so it is not decisive, but it is the right thing to cite and it cuts against the referees
χ_D	fiscal cost per household with SUD	D	0.212; CEA (2017), ≈$30,000 per person-year
G	other public spending	E	Residual that balances (20) at the benchmark, then held fixed across counterfactuals so policy is financed by τ_ℓ rather than by cutting other spending

Table 4: Income process, employment and mortality

Symbol	Meaning	St.	Value, source or target
ρ_y, σ_y	AR(1) in log productivity	L	0.90, 0.20. Storesletten, Telmer and Yaron (2004); Floden and Lindé (2001)
ε_j	deterministic age profile	D	quadratic in age, fitted to the PSID/CPS hump. **Replace with a directly estimated profile**

Symbol	Meaning	St.	Value, source or target
$\bar{\phi}_j$	baseline survival	D	**Done.** SSA period life table (`oact/STATS/table4c6`), ages 26 to 85, the two sexes combined by survivor weight $q(x) = [l^M q^M + l^F q^F]/[l^M + l^F]$ rather than averaged equally, by the mid-eighties the survivor populations differ enough that equal weights are simply wrong. Loaded from `data/us_life_table.csv`. The placeholder it replaces understated mortality badly: q at 85 was 0.045 against an actual 0.081, and at 26 0.00054 against 0.00116. This is R3 M1b
$\bar{\pi}^{ue}$	job finding, annual	L	0.55, Krueger, Cramer and Cho (2014)
$\bar{\pi}^{eu}$	job separation	C	$\rightarrow$ unemployment rate, 4.9%. BLS, ages 25+, average over the recent expansion
π_{UI}	share of entrants drawing UI	D	0.30; insured unemployment / total unemployment, FRED 2014 to 24
ϱ_{UI}	UI replacement rate	D	0.50. DOL statutory replacement rates average close to half of prior weekly wages
κ_{ueH}	employment multiplier, homeless	L	0.80, Desmond and Gershenson (2016)

Symbol	Meaning	St.	Value, source or target
κ_{ued}	employment multiplier, SUD	C	→ unemployment rate given SUD, **16.25%. Corrected, and the previous figure's stated source was wrong.** The note attributed 26.9% to a tabulation in `data/nsduh_gradients.py`; that script contains no employment variable and cannot have produced it. Measured now from the full NSDUH 2024 public use file, ages 26+, unemployment over the labour force weighted by `analwt2_c`, with `svyrdudany` for severity and `irwrkstat` for employment: 6.65% for everyone, 6.04% with no disorder, 12.72% with any drug use disorder and 16.25% with a moderate or severe one, which is the model's concept. The old target implied a relative risk of 5.49 against an overall rate of 4.9%; the data gives 2.44, so the channel was about 2.25 times too strong
κ_{uem}	employment multiplier, SMI	C	→ unemployment rate given SMI, 7.1%. NSDUH Table 6.5B: SMI prevalence 8.9% among the unemployed against 5.1% among the full-time employed, inverted by Bayes

Symbol	Meaning	St.	Value, source or target
$\kappa_{\phi H}$	mortality multiplier, homeless	D	**Now varies with age**: 4.41 at 26, 3.80 at 44, 3.14 at 56, 1.09 at 80. Baggett et al. (2013) give race adjusted all cause mortality rate ratios for homeless adults of 8.6 (men) and 9.6 (women) at 25 to 44, 4.5 at 45 to 64 and 1.1 at 65 to 84. The previous flat 1.60 was wrong twice: five times too small at prime ages, and the wrong *shape*, since a constant multiplier matching the young implies a 41% annual death rate at 80. Their ratio is the total for a population that mostly also has SUD, so it is netted down by $[(1-s)+s\kappa_{\phi d}]$ with $s=0.217$; floored at one, because the 65+ arithmetic says homelessness protects the old, which is survivor selection and not causation. §4
$\kappa_{\phi d}$	mortality multiplier, SUD	C^2	**Shape from data, level still calibrated.** → drug price participation elasticity, −0.63 (inside the −0.36 to −0.55 cocaine and −0.80 to −0.90 heroin participation range of Saffer and Chaloupka), bracketed by the 2.5 to 9.8 SMR range of Degenhardt et al. (2011). The age *profile* now comes from CDC WONDER drug poisonings, giving 5.90 at 26, 6.26 at 32, 2.62 at 56 and 1.03 at 80, while the *level* stays the second stage instrument. Flat at 6.9 it implied a sevenfold SUD mortality penalty at 80, where the measured excess is essentially nil. §4

Symbol	Meaning	St.	Value, source or target
$\kappa_{\phi m}$	mortality multiplier, SMI	D	2.50, and **constant in age, which is a finding rather than a default**. Unlike SUD and homelessness, whose multipliers collapse with age, Olfson et al. (2015) give all cause SMRs for schizophrenia of 3.6 at 20 to 34, 3.4 at 35 to 54 and 4.0 at 55 to 64: flat, if anything rising. The level is a judgement across three populations, Walker, McGee and Druss (2015) pooling 148 studies of any mental disorder at 2.22, Olfson's 3.4 to 4.0 for schizophrenia alone, and Colton and Manderscheid (2006) at about 3.2 for public mental health clients. This model's SMI state is 5.5% of the population, broader than schizophrenia and far narrower than any mental disorder, so 2.50 sits where it belongs. Sensitivity at 3.2
ρ_d	SUD productivity penalty	D	1.00 in the baseline: no direct wage effect. The employment-margin evidence is robust (about 50% more unscheduled absence, 44% higher turnover) and is carried by κ_{ued}; the wage penalty conditional on working is contested, with parts of the older literature finding positive effects for casual use. A version calibrated to the SUD income ratio is an **appendix robustness** (AE A20)
ρ_m	SMI productivity penalty	C	$\rightarrow$ homeless income / average, 19.1%. Meyer et al. (2025), homeless population linked to administrative earnings records

Table 5: Addiction, recovery, treatment and mental illness

Symbol	Meaning	St.	Value, source or target
π_d	addiction hazard given any use	C	→ SUD rate, 3.68%. NSDUH 2024 microdata, ages 26+
j_E	maximum experimentation age	S	Real age 55, with no source, R1's smaller comment 1 says so. Run with and without (§7); if it can be dropped it should be
$\bar{\pi}_r$	baseline recovery hazard, housed	D	0.025/yr; Vaillant (1995). The AE questions applying alcohol recovery rates to drug SUD; Fleury et al. (2016) give 35 to 54% remission at a mean 17-year follow up across substances, which brackets this
μ_H	recovery multiplier, homeless	C	→ drug use disorder share of the homeless, 21.7%. Gutwinski et al. (2021), *PLOS Medicine*, 39 pooled studies. Note this is the DRUG figure; the submitted paper's 37% is their ALCOHOL figure
μ_m	recovery multiplier, SMI	D	**0.82.** Dual-diagnosis cohorts relapse at 39.8% against 26.4% without a co-occurring disorder (odds ratio 1.85 for difficulty maintaining abstinence), so sustained remission runs 0.602 against 0.736, a ratio of 0.82; converting the odds ratio to a risk ratio at that base rate gives the same figure. Previously 0.50, which was invented
g	treatment effect on recovery	S	1.35, from an odds ratio of 1.22 for contingency management in the only clean randomised evidence. Specialty-treatment remission of 47.6% and overall remission of 35 to 54% are *not* treatment effects, different populations, different selection, which is why they are not used here. **The TF result runs through this**, report a range and the value at which the ranking flips

Symbol	Meaning	St.	Value, source or target
p_m	SMI prevalence	C	$\rightarrow$ 5.5%, NSDUH 2023, adults. A target rather than a parameter now, since onset varies with SUD
λ_m	SMI recovery rate	D	0.067/yr; İmrohoroğlu, Zhao's own moment, ≈50% improve enough to work within 10 years
onset_m	SMI onset	C	$\rightarrow$ SMI prevalence, 5.5%. No longer derived, because onset is no longer common to everyone
ι	SUD multiplier on SMI onset	D	**2.0, new.** Danish registers: any substance use disorder predicts conversion to schizophrenia with a hazard ratio of 1.34, amphetamine use disorder 1.90, opioid use disorder 2.74; substance-induced psychosis converts at a pooled 25% (Murrie et al.). Range 1.3 to 2.7
ζ_{DI}	disability coverage given SMI	C	$\rightarrow$ SMI share of the homeless, 25%. Gutwinski et al. (2021): 12.4% schizophrenia-spectrum plus 12.6% major depression. This is İmrohoroğlu, Zhao's own identifying margin (ζ_{DI} in their notation)
$\bar{y}_{DI}$	disability benefit level	D	0.25 of average income. SSA: average SSDI benefit $1,537 a month in 2024, about $18,400 a year against average annual earnings near $70,000, so 0.26. The SSI federal benefit rate of $943 a month is 0.16; SSDI is the relevant figure for people with a work history
ψ_1	drug supply shifter	N	Normalised so $p_v = 1$ at the benchmark quantity, which is what lets the price be endogenous without disturbing the existing normalisation (R3 M2d)

Symbol	Meaning	St.	Value, source or target
ψ_2	drug supply curvature	D	Elasticity $1/(\psi_2 - 1)$, **baseline 2.0**, bracketed from measured tax incidence in two legal markets. **Lower bound 0.79**: Hansen, Miller and Weber (2018) on Washington's 2015 switch from a 25% gross receipts tax at every supply chain stage to a single 37% retail excise, in universal state traceability data. Consumers bear 44% of the additional burden and demand elasticity is about -1 within two weeks, so $\varepsilon_s = \rho\lvert\varepsilon_d\rvert/(1-\rho) = 0.79$. Washington caps licensed cultivation canopy, a constraint with no analogue in an illegal market, so this is a floor. **Upper bound about 3.6**: the same paper notes its result stands in contrast to the cigarette and gasoline incidence literatures, which find consumers bear the substantial majority; at a consumer share near 0.9 and a consensus demand elasticity near 0.4 that implies about 3.6 for unconstrained legal manufacturing. The two cannot be reconciled by demand elasticity alone, which is the authors' own explanation, since no single supply elasticity fits both, so cannabis genuinely is steeper and the licence cap is why. **Illegal drugs sit between**: no licence cap pushes toward 3.6, while enforcement prevents capital intensive scaling, drugs being "diluted and packaged by hand ... because enforcement makes it risky to maintain dedicated, fixed capital equipment" (Caulkins and Reuter 2010), and moving more product needs more shipments and couriers, pushing toward 0.79. Swept over $\{0.79, 2, 5, 50\}$, where 50 is the effectively infinite benchmark this model previously assumed. **This makes the equilibrium price**

Symbol	Meaning	St.	Value, source or target

Superscript 2 marks the two parameters calibrated in the *second* stage, against experimental rather than cross-sectional moments.

Parameters that remain undisciplined. Five, and they are flagged rather than hidden: g (treatment effect), μ_m, $\bar{y}_{DI}$, ψ_2 and κ_x. Of these only g carries a headline result, and it is reported as a range. The baseline survival profile $\bar{\phi}_j$ is currently a smooth functional form and should be replaced with US life tables before the paper is resubmitted.

4 Hazards that vary with age

Four hazards were constants and should not have been. Each was calibrated or assigned to hit an average, and each therefore matched a level while getting the life cycle profile wrong, in every case in the direction that weakens the paper's own mechanism at the ages where it does its work.

Baseline survival. Was a smooth exponential fitted to nothing. Now the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$, rather than averaged equally: by the mid eighties the two survivor populations differ enough that equal weights are simply wrong. The placeholder understated mortality by nearly half at the top of the range, $q = 0.045$ at 85 against an actual 0.081. R3 M1b.

Homelessness and SUD mortality, derived jointly. These cannot be set separately. Baggett et al.'s homelessness ratio is measured on a population that mostly also has SUD, so netting the SUD component out depends on wherever the SUD multiplier lands:

$$\kappa_{\phi d}(a) = 1 + (\kappa_{\phi d} - 1)\,\sigma_d(a), \tag{21}$$

$$\kappa_{\phi H}(a) = \max\Big\{1,\ \frac{RR(a)}{(1-s) + s\,\kappa_{\phi d}(a)}\Big\}, \tag{22}$$

where σ_d is the SUD profile from CDC drug poisonings normalised to one at its peak, RR is Baggett's all cause ratio, and $s = 0.217$ is the model's own SUD share of the homeless. Because $\kappa_{\phi d}$ falls with age, so does the correction: at 45 to 64 more of the observed ratio is genuine homelessness effect, and $\kappa_{\phi H}$ at 56 comes out at 3.14 rather than the 1.97 a flat calculation gives.

Why the level stays free. $\kappa_{\phi d}$ is the second stage's only instrument: it is bracketed inside the Degenhardt SMR range to hit a drug price elasticity of participation of -0.63, and that elasticity is what disciplines how much drug use responds to its costs, which is the object the entire policy experiment turns on. Pinning the level from mortality data would take the instrument away and leave the elasticity undisciplined. So the data supplies the *profile* and the elasticity supplies the

level. The system is then over identified, which is the point: §6 reports the age weighted SMR the model produces against the published ratios, and a disagreement is a finding rather than something a dual purpose parameter can hide.

SMI onset. Was a scalar. A flat onset against a flat recovery drives prevalence toward its steady state from below, so the model produced a profile that *rises* with age; NSDUH has SMI falling from 11.0% at 26 to 29 to 1.4% at 65 and over, a factor of eight. The model hit the 5.5% average and had the shape backwards, which misplaces the mental health channel and everything it feeds. Onset is backed out of the prevalence profile and the recovery hazard, $\lambda(a) = [p(a{+}1) - p(a)(1-\mu)]/(1-p(a))$, with prevalence fitted log linearly ($R^2 = 0.977$) rather than interpolated piecewise: an early attempt differenced a piecewise linear curve and every kink became a spike, giving an onset profile that was 1.00 at 26, 0.01 at 32 and exactly zero at 56 and 68. Those zeros were artifacts, not incidence that vanishes and reappears. As with the mortality multipliers the shape is data and the level stays calibrated to the prevalence target. Newborns now enter with the observed prevalence at 26 rather than the ergodic of the age one transition matrix, which would start them at 4.7% against 11 to 13% observed.

What stays constant, and why. $\kappa_{\phi m}$. I expected it to collapse with age like the other two and it does not: Olfson et al. (2015) give all cause SMRs for schizophrenia of 3.6, 3.4 and 4.0 across 20 to 34, 35 to 54 and 55 to 64. Flat, if anything rising. The employment hazards π_{ue}, π_{eu} also stay constant; job finding does fall with age in the data, but it shapes the income profile rather than any of the paper's mechanisms, and it is not worth the extra machinery.

5 Calibration

Two stages, and they are different exercises. The first matches a cross-section of the stationary equilibrium. The second matches short run *responses* measured in experiments, which cannot be computed by comparing stationary equilibria.

5.1 First stage: the stationary equilibrium

Nineteen parameters are calibrated jointly against twenty-one moments of the stationary distribution. All the assignments are in Section 3; what matters methodologically is the following.

The parameter, moment map is close to triangular, and the code says so. Each parameter has one moment it dominates, and the sign of $\partial M/\partial\theta$ is declared explicitly for each pair, so a mistaken sign shows up immediately as a moment walking away from its target rather than as a silent failure.

Three moments are step-like and must be bracketed, not stepped. Participation is a near-threshold function of b_0, below a cutoff everyone uses, above it nobody does, and in between every drug moment reads zero. Homelessness moved from 0.12% to 3.29% for a 35% change in

h_{HL}. Wealth/GDP is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. A proportional step of 0.9 takes a discount factor from 0.96 to 0.864, which is enormous in a life cycle model; the first attempt at this calibration drove β into its lower bound and took every other moment with it. All three are bisected at the head of each sweep, in partial equilibrium, before anything else moves.

The discount factor and wealth/GDP. β targets aggregate wealth equal to three times GDP, with GDP measured as *all factor income*, labour earnings plus the return on the capital stock, and transfers excluded, because transfers are redistribution rather than output. Three is the standard US ratio of the private capital stock to annual output. It is the right target here rather than the SCF's net-worth-to-*household-income* ratio of roughly 7.3, whose numerator is mostly housing equity a model without homeownership cannot hold and whose denominator is not output. One implication should be stated in the paper rather than left for a reader to find: with $A = 3 \cdot \text{GDP}$ and GDP = labour income $+ rA$, capital income is 9% of GDP, low against a one-third capital share. That follows from holding r fixed at 3% in a model with no production function.

Three targets that were wrong in the submitted paper.

1. *SUD among the homeless.* The paper targets 37%. Gutwinski et al. (2021) pool 39 studies of homeless people in high income countries and report 36.7% *alcohol* use disorder and 21.7% *drug* use disorder. The model is about drug use. This is the AE's A16.

2. *Income of drug users.* Referee 1 (2c-i) attacked the model's 41% against "roughly 74%". The 74% is computed over people with a use *disorder*, not people who use. For ages 26+, NSDUH gives 95.1% for users and 68.7% for moderate or severe use disorder. Targeting the disorder ratio and *validating* on the use ratio is the sharper test, and it makes the economics explicit: use is close to income-neutral, and it is progression to disorder that concentrates at the bottom.

3. *Housing burden.* Everyone rents in this model, so a renters-only target would import the selection into who rents. The target is the aggregate ratio over *all* households with owner housing valued at imputed rent, and the full distribution is used for validation.

5.2 Second stage: short run experimental moments

Method note. These are partial equilibrium **cohort** experiments, not transition paths. A trial of two hundred people does not move the tax rate or the drug price, so prices are held at their benchmark values; the household problem *is* re-solved, because each arm changes the rules a forward-looking household faces. Only the aggregate policy counterfactuals of R1 3(d) need a transition solver.

Referee 2 (C1): "all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access". Referee 3 and the AE (A13) say the same independently. Referee 1 (2d, 2e) asks for policy-relevant elasticities from the RCT literature and notes that the price-elasticity moment as written is not replicable.

These are short run objects. A trial hands a treatment to a selected group and measures them months later; it does not move aggregate prices and it does not wait for a new stationary distribution. The second stage holds prices at benchmark, takes the benchmark distribution as

the initial condition, restricts to the trial's eligible population, applies each arm's rules, re-solves the household problem, and pushes the cohort forward.

The housing-access elasticity. Milby et al. (2005) randomised 196 cocaine-dependent homeless adults into three arms, all receiving identical day treatment:

Arm	Model counterpart	n	Weeks abstinent (of 24)
NH	no assistance	66	4.48
NACH	voucher, unconditional, *Housing First*	67	6.60
ACH	voucher, conditional, *Treatment First*	63	8.32

The targeted moment is the *ratio* NACH/NH = 1.473, which identifies ρ_H. Ratios rather than levels: "mean consecutive weeks abstinent out of 24" is not the same object as "abstains this model year", but the treatment effect is comparable and it is the treatment effect the referees want reproduced.

The conditionality effect is a validation check, not a target. The abstract reports that the abstinence-contingent arm beat the unconditional arm, but only *after controlling for treatment attendance*. In the raw randomised comparison NACH versus ACH gives $P = .091$, and the GEE models state the two housed groups "did not differ from one another". Calibrating to a point estimate the source reports as insignificant is exactly the error Referee 1 attacked κ_{cons} for.

The extensive/intensive contrast is the sharper test. Milby's counterfactual is literally no housing, and the effect is large. The Housing First RCTs compare Housing First to "usual services", which already include shelter or transitional housing, and the systematic review finds "no clear differences ... on outcomes related to mental health, quality of life and substance use". These are not in conflict: the elasticity is *large at the extensive margin* and *near zero at the intensive margin*. A model reproducing both is far better identified than one matching either alone.

The drug-price participation elasticity. Perturb p_v, hold everything else at benchmark, re-solve, read the arc elasticity of participation. Partial equilibrium deliberately, since the empirical estimates come from price variation rather than from a comparison of steady states. Target -0.63; identifies $\kappa_{\phi d}$ within the Degenhardt SMR range. Referee 1 also asks for robustness across -0.36 to -0.9, which is the same bisection at the two endpoints.

The value of a statistical life. Reported rather than targeted, because $\aleph$ currently carries a first stage moment. Reference \$7.5m (Viscusi and Aldy 2003; EPA 2010). If the model implied value is far off, the right response is to move $\aleph$ to the VSL target and reassign the SUD income ratio to ρ_d.

6 Validation

New untargeted moments. None of these is fitted, and each is produced endogenously by a model with sixty single ages.

Moment	Data	Source
Homelessness rate, 25 to 34 / 35 to 44 / 45 to 54 / 55 to 64 / 65+	0.316 / 0.338 / 0.291 / 0.250 / 0.077%	H
Shares of the homeless across those bands	26.0 / 27.2 / 21.0 / 18.4 / 7.5%	
Returns to homelessness within 24 months	17.4%	HUD SPM 2024
Drug poisonings as a share of all deaths, 25 to 44	about 20%	CDC WONDER
Bottom 50% share of net worth	2.52%	Fed DFA, 2022 t
Housing cost above 30% / 50% of income	40.6 / 17.5%	SCF 2022
Age weighted SMR, SUD	2.5 to 9.8	Degenhardt et al
Age weighted SMR, homeless	9.1 / 4.5 / 1.1 by band	Baggett et al.
Tenant contribution rate	31.2%	HUD, tenant bas
Voucher holders with wages as main income	28%	HUD
SUD income / average, moderate or severe	71.1%	NSDUH 2024
Labour force participation, SUD vs clean	65.8 vs 61.5%	NSDUH 2024
Homeless at treatment admission	14.5%	TEDS-A 2022
Co-occurring psychiatric problem at admission	43.1%	TEDS-A 2022
Working at all, among the homeless in treatment	6.6%	TEDS-A 2022

The age weighted SMRs deserve emphasis. Because the multipliers now vary with age while the literature reports a single ratio, the comparable object is the SMR computed on the model's *own* age distribution of SUD and of homelessness. An unweighted average across ages would compare a profile against a statistic measured on a population concentrated at its young end, where both multipliers are largest. This is reported automatically by the second stage.

None of these is targeted. Referee 1's 2(a), model fit never reported, and the AE's A12 are answered by reporting all of them, including the misses, and by framing the misses honestly rather than favourably (2b).

Table 6: Untargeted moments

Moment	Data	Source
Distributional		
Full CDF of housing cost / income	12 thresholds, 5 to 100%	SCF 2022, ages 26 to 85, owner housing at imputed rent
Income of drug *users* / average	95.1%	NSDUH 2024, ages 26+
Income of those with SUD / average	68.7% (targeted; reported alongside)	NSDUH 2024, ages 26+
Share of the homeless employed	40 to 52%	R1 comment 2(c)(i)
Age distribution of the homeless	48.1% aged 26 to 44; 27.1% 55+	R1 comment 2(c)(ii); to be reported in finer bins
Drug use by age	U-shaped	NSDUH microdata
Wealth by age	,	SCF 2022

Moment	Data	Source
Policy responses		
HF effect on 3-year mortality	−4.6pp (s.e. 1.7)	Costantini (2025), ≈300,000 VA veterans, IV
TF effect on long run health	≈ 0	Costantini (2025). The outcome is mortality and health-service use; the model captures mortality and SUD status, not general morbidity
Housing stability, HF vs TF	sustained vs declining	HF RCT literature
PSH vs emergency shelter, property values	PSH raises, shelter lowers	shelter-siting hedonic literature

The Costantini result is the paper's largest single risk and belongs in the introduction. The best available quasi-experimental evidence says the mortality benefit runs through *housing*, and that conditioning on treatment adds nothing detectable. The submitted paper concludes universal Treatment First dominates. The defensible position after the revision is the *generosity* result, expanding coverage dominates, conditionality is second-order and uncertain, which is consistent with all of the above and is still a paper. Getting there requires reporting welfare and mortality as two dimensions, the AE's A9, rather than collapsing them into one ranking.

7 Diagnostics the model must pass

Two checks are run before any number is read as a result.

Is life worth living? Death is worth zero in (19). If any household has $V_j(s) < 0$ it strictly prefers to die, and for that household drug use is *rewarded* by the mortality it causes, inverting the paper's central mechanism for exactly the population the policy targets. The diagnostic reports the minimum V over the benchmark distribution, the mass with $V < 0$, and whether they are disproportionately homeless, addicted or retired. If any mass appears, $\aleph$ is too small.

The age profile of use, with and without j_E. Referee 1's smaller comment 1 says the maximum experimentation age is in tension with the U-shaped age profile of use. The test is whether removing it produces a sensible profile or piles use onto the old, who have the least remaining life to lose. If use does concentrate at old ages, that is informative rather than embarrassing, it says the deterrent is the value of remaining life, but it has to be reported and explained rather than assumed away by a hard age limit.

8 Mechanisms: shutting channels and recalibrating b

8.1 Why this replaces the static appendix

Referee 1's comment 1(d) does not merely ask for intuition, it *specifies the design*: show that the mortality channel is the quantitatively important one by shutting channels off. Comment 1(e) says the current appendix is disorganised, static and overreaches. Referee 2 (C4) wants the drug-use policy function shown by income and wealth. Referee 3 (M2b) asks what patterns the Stone–Geary specification actually generates and where heterogeneity in v comes from; (M2c) asks the sharpest theoretical question in any report, since drug use is rational and forward-looking, where do the externalities justifying intervention arise? The AE (A1) asks for a stylised section because the number of moving parts obscures the economics.

One exercise answers all five, run on the *calibrated* model rather than a separate static toy. That matters: a static model with no assets cannot address Referee 2's question about wealth, and a toy calibrated to nothing cannot establish that a channel is quantitatively important.

8.2 The design

Shut one channel at a time and, in each arm, **recalibrate b_0 so that drug participation returns to its benchmark value**. The recalibration is what makes the comparison informative: without it, removing the mortality penalty simply raises use everywhere and the level change swamps any compositional change. Holding participation fixed separates *sorting* from *level* and yields two statistics per arm:

Statistic	What it measures
How far b_0 had to move	how much deterrence the channel was providing
The income gradient of use	*whom* it deterred

The arms run in partial equilibrium at benchmark prices: the object of interest is the behavioural mechanism, not equilibrium feedback, and holding prices fixed isolates it as well as being far cheaper.

8.3 The arms

1. **Benchmark.** All channels on.
2. **No mortality penalty**, $\kappa_{\phi d} = 1$. The paper's headline claim.
3. **No employment penalty**, $\kappa_{ued} = 1$. Referee 1's rival hypothesis.
4. **No homelessness feedback**, $\kappa_{\phi H} = \kappa_{ueH} = \mu_H = 1$ and $\rho_H = 1$. Isolates the loop that makes homelessness and SUD mutually reinforcing.
5. **All private costs off at once.** Whatever deterrence survives is *resource diversion*: the budget constraint alone. Diversion cannot be switched off directly, drugs are priced inside a Cobb–Douglas bundle, so making them free sends v to the corner, but it is measurable as the residual, which is what this arm reports.

8.4 What is reported

Table 7: Mechanism decomposition (structure; numbers pending the calibration)

	Benchmark	No mortality penalty	No employment penalty	No homeless feedback	All private costs off
b_0 relative to benchmark	1.00				
Participation (%, held fixed)					
SUD rate (%)					
Homelessness (%)					
Use rate, income Q1 (%)					
Use rate, income Q5 (%)					
User income / average					
experimenters only					
SUD only					
Users unemployed (%)					
Users homeless (%)					
Mean age of users					
Mean assets of users					

8.5 The sharp prediction, and how it could fail

With preferences fixed across addiction status, becoming addicted no longer relaxes any constraint, so *all* deterrence must come from mortality, employment and the budget. If removing the mortality penalty requires only a small change in b_0 and leaves the income gradient nearly flat, the paper's central mechanism is not carrying the weight the text claims, and the honest response is to say so and re-centre on the generosity result. If b_0 must fall substantially and the gradient inverts, the mechanism is established numerically on the calibrated model, which is what Referee 1 asked for and what the static appendix could not deliver.

8.6 Where the externalities come from

Referee 3's M2c asks where the externalities justifying intervention arise, given that use is rational and forward-looking. They are of two kinds.

Fiscal, and it has five parts. A household that acquires SUD (i) works less and pays less tax; (ii) becomes likelier to be voucher-eligible and to receive one; (iii) becomes likelier to be homeless, which raises the direct programme cost *and* makes work harder to find and easier to lose ($\kappa_{ueH} = 0.80$, Desmond and Gershenson), compounding back into (i); (iv) becomes likelier to develop serious mental illness, which raises disability receipt and feeds back into employment, mortality and homelessness; and (v) dies earlier, truncating the whole stream of future revenue. None of this is internalised, and all of it is already in the model, but the submitted paper never reported it, and did not have channel (iv) at all.

It is now measured. Define the expected present discounted value of a household's net contri-

bution to the government,

$$N_j(s) = \underbrace{T_j(s)}_{\text{taxes}} - \underbrace{Vch_j(s)}_{\text{vouchers}} - \underbrace{Tr_j(s)}_{\text{other transfers}} - \underbrace{\chi_H \mathbf{1}\{h' = h_1\} + \chi_D \mathbf{1}\{d = 1\}}_{\text{direct costs}} + \frac{\phi_j(s)}{1+r}\,\mathbb{E}\big[N_{j+1}(s')\big], \quad (23)$$

discounted at the government's rate. Given the benchmark policy functions this is a *linear* recursion, so it costs one backward pass and no optimisation, and carrying the four components separately decomposes the answer. The fiscal cost of one more case is

$$\mathbb{E}\big[N(a, y, e, d = 0, m, \eta)\big] - \mathbb{E}\big[N(a, y, e, d = 1, m, \eta)\big]$$

over the benchmark distribution of clean households: the same household, counterfactually with and without SUD.

Two of the four channels are separated by re-running the recursion with one transition altered. Evaluating the SUD stream at *clean* survival isolates what early death alone costs, purely by truncating the revenue stream. Evaluating it with homelessness carrying *no* employment penalty isolates the SUD $\rightarrow$ homelessness $\rightarrow$ unemployment $\rightarrow$ lost-tax chain. Both are easy to miss precisely because they work through transitions rather than through any flow.

Reporting them in one currency. The fiscal loss has to be financed by somebody, so annuitising it and spreading it across the population puts it in the same units as the disamenity, a share of average annual consumption. Only then can a reader see which channel actually matters. On the target values alone the direct outlays $\chi_H H + \chi_D D$ come to about 0.86% of average income per capita against a disamenity near 0.33%, and the direct outlays are only one of the four fiscal channels. If that ordering survives the calibration it is a substantially stronger position than the submitted paper's, where the entire ranking rested on κ_{cons}: the case for intervention would rest mainly on the channel that is measured inside the model and needs no free parameter, rather than on the contingent-valuation number, which is the weakest link.

A second, wider measure. The present-value recursion above is exact but partial: it prices what the government gains and loses, not what *households* gain and lose, so it misses three things. It misses the deadweight loss of financing the transfer, which is real and first-order here because labour supply is elastic. It misses the drug price, which is an equilibrium object, so one more user raises the price every other user pays. And it misses the transfer scale: average income sets the food-stamp floor and the disability benefit, median income sets voucher eligibility, so one more case of SUD lowers all three for everybody. The wider measure raises the addiction hazard π_d by ten per cent, re-solves, and reports the change in newborn welfare as a consumption-equivalent variation. Each channel is then isolated by re-solving with that channel alone *frozen* at its benchmark value; the gap between the free re-solve and the frozen one is that channel's contribution.

Two cautions on reading the resulting table. First, the total includes the household's own expected cost, which it internalises and which is therefore *not* an externality; only the channels beneath it are. Second, the disamenity, drug-price and transfer-scale channels are each measured with the other channels live, so they are marginal contributions and need not sum to the total, the sum is a diagnostic, not an accounting identity. The fiscal pair is the exception: the mechanical

transfer is measured by holding hours at their benchmark schedule while the tax moves, so no distortion is created, and the deadweight loss is the residual once hours are free to respond. Those two are nested by construction and do add up. Holding hours fixed has to be done by constraining the choice, not by reweighting leisure in preferences: a large weight on leisure does not fix hours, it drives them to the corner.

Non-fiscal. The disamenity term in (1). Referee 2 (C2) asked for it and said it was easy to introduce; it is, but it has no credible point estimate. Hedonic shelter-siting studies measure a *local, avoidable* disamenity, what people pay to live further away, and this model has no geography, so nobody can avoid anything and a capitalisation coefficient does not map onto an aggregate public bad. Corbae, Glover and Nattinger's externality is about neighbourhood housing *quality* rather than homelessness, and their identification comes from rent decontrol, so what transfers is their identification logic, not their functional form.

A calibration for the disamenity does exist. Referee 2 noted that introducing the externality "would require a new calibration moment". There is one, and it is a much better fit to this model than the hedonics. Loubière et al. ran an eight-country contingent-valuation study (France, Ireland, Italy, Netherlands, Poland, Portugal, Spain, Sweden, roughly 700 respondents each, fielded in 2017) asking exactly the right question: what would you pay in annual taxes to *end* homelessness through Housing First? The answer was about €57 per respondent per year, roughly 0.2% of average annual earnings. That is a stated preference over the *aggregate* homelessness rate, precisely the object κ_x multiplies, and it is non-spatial, which the hedonic estimates are not.

The normalisation that this requires. The €57 buys elimination of the *European* homelessness rate, not the model's. Dividing by the model's own rate would make the same survey figure imply a different disamenity in every economy it was transplanted into: mechanically small wherever homelessness is rare and large wherever it is common, which is the opposite of what a per-unit valuation means. So κ_x is set as a price per unit of homelessness,

$$\kappa_x = \frac{\mathrm{WTP} \cdot \mathbb{E}\big[c\, u_c\big]}{H_{\mathrm{ref}}\big(1 + \omega_x s_{\mathrm{ref}}\big)}, \tag{24}$$

with H_{ref} the European rate at the time of the survey and s_{ref} the drug-use-disorder share of the homeless in the same pooled evidence base. FEANTSA counted roughly 700,000 people sleeping rough or in shelter across the EU in 2017 against a population near 440 million, so $H_{\mathrm{ref}} \approx 0.16\%$; the 2023 overview reports about 895,000, but part of that rise is improved measurement. The central value used is 0.18% and the range swept is 0.15 to 0.22%. Since κ_x is inversely proportional to H_{ref}, this is the single most consequential judgement in the disamenity calibration and it must be reported as a range. At a model homelessness rate of 0.30% against a reference of 0.18%, the adjustment scales the disamenity by 1.67, from 0.20% to 0.33% of income.

Caveats, to be stated in the paper. Stated rather than revealed preference; European rather than US; 30.3% protest zeros; and the elicited change is elimination rather than a marginal change, so the linear extrapolation to the model's rate is an assumption.

What the hedonics are still good for. Not a magnitude, but a sign. The NYC Independent Budget Office finds a property within 500 feet of a family shelter sells for about 6.4% less than

a comparable property 500 to 1,000 feet away, and properties near two or more shelters 17.4% less than near one. The Furman Center, studying 123 supportive-housing developments over 18 years, finds supportive housing does *not* depress neighbouring values and slightly raises the closest ones; spatial difference-in-differences work reaches the same split. Permanent supportive housing is Housing First-like and temporary congregate shelter is Treatment First-like, so the externality plausibly differs in *sign* between the two policies being compared. That is a substantive point in Housing First's favour and it does not require a number.

Note finally that $\kappa_x = 0$ is a defensible baseline in a way $\kappa_{\text{cons}} = 0$ was not: that was the value at which the paper's conclusion reversed.

9 What each referee still wants

9.1 What has been done, and what it answers

Table 8: Model changes and the comments they answer

#	Change	Answers
1	Homeownership, tenure, mortgages and negative assets removed	AE A4; R2 D1
2	Housing market GE removed	AE A5
3	Drug price made an equilibrium object, ψ_1 normalised	R1 1(c); R3 M2d; AE A5
4	Cobb–Douglas weights fixed across SUD status	R2 C5; R3 M2a; AE A8
5	κ_{cons} deleted; homelessness externality in period utility, weighted by composition	R1 3(a), 3(b); R2 C2; R3 M2c; AE A9
6	Endogenous treatment take up, at a utility cost, with a take up target	R1 1(b); AE A10
7	Imperfect detection and false positives	R1 4(a), 4(b); AE A11
8	Serious mental illness, six channels	R1 1(a), 2(c)(ii)
9	SUD wage penalty set to one in the baseline, with the evidence stated	AE A20
10	UI drawn on entry, one year; four states collapse to three exactly	AE A2
11	Estates to the government rather than lump-sum by type	AE A7
12	Single addiction hazard; $\bar{v}$ and the 6.48 ratio gone	R3 M2b (partly)
13	Continuous labour supply from the FOC, welfare trap handled exactly	R3 m1; AE A3
14	β retargeted to wealth/GDP = 3	AE A17
15	Rent normalised; the whole housing grid calibrated	R2 D1

#	Change	Answers
16	Homeless SUD target corrected 37% $\rightarrow$ 21.7%	AE A16
17	Drug user income target split: users 95.1%, SUD 68.7%	R1 2(c)(i)
18	Housing burden targeted over all households, full CDF for validation	R1 2(a); AE A12
19	Every parameter given a source or a target (§3)	AE A17
20	Capital income taxed in the household budget; true median income	internal consistency
21	Mechanism ladder built	R1 1(d), 1(e); R2 C4; R3 M2b, M2c; AE A1
22	Baseline survival from the SSA life table, sexes combined by survivor weight; the fitted exponential is gone	**R3 M1b, closed**
23	χ_H from Abramson and Van Nieuwerburgh, \$2,500 per household month for the literally homeless, giving 0.4286	**AE A15, closed**
24	Housing grid identified from the ACS B25042 size distribution rather than from housing cost burdens, which become untargeted validation	**R2 D1, strengthened**
25	Homelessness target corrected to a household basis, 0.399%	**internal consistency**
26	Homelessness and SUD mortality multipliers made age varying and derived jointly; SMI onset likewise; $\kappa_{\phi m}$ shown to be flat	**R1 2(c), R3 M1b**
27	Drug testing costed in the government budget; detection requires a test	**R1 4(c), closed**
28	b_1 given a target, the Milby abstinence rate among the homeless with SUD	**no longer asserted**
29	Treatment as a *utility* cost validated against NSDUH: the three largest barriers are beliefs and preferences, not prices	**R1 1(b), AE A10**

9.2 Referee 1's rival hypothesis, and what the data did to it

R1's central alternative is that the SUD to employment channel, rather than mortality, does the work. The model was biased toward that reading on *both* sides at once, and neither bias was

visible until the underlying numbers were measured rather than assumed.

Channel	Was	Is	Source
Unemployment given SUD	26.9%	**16.25%**	NSDUH 2024 microdata
implied relative risk	5.49	**2.44**	
SUD mortality multiplier	6.9, flat	profile, peak 6.9	CDC WONDER
Homelessness mortality multiplier	1.6, flat	4.41 at 26, 3.14 at 56	Baggett et al.
Baseline survival	fitted exponential	SSA life table	q_{85} 0.045 $\rightarrow$ 0.081

So the employment channel was roughly 2.25 times too strong and the mortality channel was running at something like a fifth of its measured strength at prime ages. Correcting both moves against R1, and from measurement rather than assertion, which is the part that matters for a response letter.

Two further facts from the same file are worth reporting because nobody had looked at them. Mean personal income among those with a moderate or severe drug use disorder is 71.1% of the average at ages 26+, against a 68.7% target the model was already fitting, so that moment is nearly hit unprompted. And labour force *participation* is slightly HIGHER among those with a disorder, 65.8 against 61.5%: the penalty runs through unemployment conditional on participating, not through dropping out of the labour force. The model should reproduce that, and it is a sharper statement of the mechanism than the paper currently makes.

9.3 Referee 1, outstanding

#	Comment	Status
1(d),(e)	Show the mortality channel is the quantitatively important one; reorganise the appendix	Ladder built, **not yet run**, needs the calibration
2(a)	Model fit never reported	**Pending the calibration**
2(b)	Tables 3 to 4 misses framed too positively	**Writing**, once the misses are known
2(c)(i)	Homeless 100% unemployed vs 40 to 52%; income $\approx$ 0 vs 19.1%	Mechanism added; the employment share is **not yet a targeted moment**
2(c)(ii)	Homeless too young	SMI added; **age distribution not yet computed**, and needs finer bins
2(d)	Match policy-relevant RCT elasticities	**Second stage built, not yet run**
2(e)	Price-elasticity moment not replicable	Model counterpart defined; **appendix not written**
2(f)	Show identification numerically	**Not started.** One solve per parameter

#	Comment	Status
3(c)	Tone down; universal TF not unambiguously preferred	**Writing**
3(d)	Solve proper transition paths	**Not built.** The critical path
4(c)	Compliance, stigma and administrative costs of testing	**Not started.** Needs a real admin cost in the budget
4(d)	Sensitivity ladder over addiction, recovery, mortality, employment	**Not started**
sm. 1	Why a maximum experimentation age?	Now switchable; **diagnostic not yet run**
sm. 2 to 10	Levels and percentage changes; typographical and citation fixes	**Writing**

9.4 Referee 2, outstanding

#	Comment	Status
C1	The elasticity of drug use to housing access	ρ_H added and the moment defined; **not yet calibrated**
C2	Homelessness externality	Implemented, $\kappa_x = 0$; **no credible calibration**, see §8.6
C3	Parental anxiety about children	**Open decision.** R2 says it "could be discussed"; a warm-glow term would answer this and AE A6 together
C4	Show the drug-use policy function by income and wealth	**Not produced.** Free once calibrated
C6	Optimal voucher design; is χ_H too high?	**Not started.** One sweep of π_V plus a χ_H sensitivity

9.5 Referee 3, outstanding

#	Comment	Status
M1a	Stylised facts on drug use, voucher receipt and homelessness	**Not written**
M1b	Empirical basis for survival and unemployment hazards	Partly done in §3; $\bar{\phi}_j$ is still a placeholder and must come from life tables
M1c	Show the empirical income, drug-use relationship	Microdata extracted; **figure not made**

#	Comment	Status
M2b	What patterns does Stone–Geary generate?	Ladder built, **not run**
M2c	Where do the externalities arise?	Enumerated in §8.6; the fiscal side needs quantifying from the ladder
m1	Labour supply and the welfare cost of tax financing	Labour supply now continuous; **welfare decomposition not done**
m2	Sensitivity as graphs, not tables	**Not started**
m3	Vary voucher receipt continuously; Laffer curve?	**Not started**

9.6 Associate Editor, outstanding

#	Comment	Status
A1	Stylised model section	Plan is to use the calibrated model plus the ladder; **not written**
A6	Intergenerational language without intergenerational linkages	**Open decision**: delete the language, or add a warm-glow term (with R2 C3)
A9	Report welfare and mortality as two dimensions	**Not done.** Requires the counterfactuals
A12	Model fit; non-targeted moments	**Pending the calibration**
A13	Key elasticities	**Second stage built, not run**
A14	Connect to Costantini, Cohen, Brounstein and Wieselthier, Abramson, Humphries et al.	**Writing**
A15	Compare χ_H with Abramson and van Nieuwerburgh	**Done**, §3. Their \$1,775/month blended figure implies $\chi_H = 0.304$ and their literally-homeless \$2,500 implies 0.429, both above the 0.252 the referees called too high
A16	Homelessness definition	Target corrected; the paper must still state that it targets *literal* homelessness and excludes doubling-up
A17	Rewrite the calibration section	§3 to 5 are the raw material; **not yet in the paper**

#	Comment	Status
A18	Fewer counterfactuals in the body	**Writing**
A19	"Addicted" → "suffering from substance use disorder"	**Writing**, global replace

9.7 Loose ends, split by status

Almost everything in the four lists above is closed by one of four blocks of work already scoped: the **recalibration**, the **counterfactuals**, the **mechanism ladder** and the **rewriting**. What is left is sorted here into what is finished, what is decided and scoped but not yet built, and what is genuinely unresolved. Nothing sits in the third table merely because it is unfinished.

Ref	Item	What closed it
R3 M1b	Survival hazards	SSA period life table, ages 26 to 85, sexes combined by survivor weight. The fitted exponential understated mortality by nearly half at 85
R1 2(c)	SUD and homelessness mortality	Both multipliers now vary with age and are derived jointly, from CDC drug poisonings and Baggett et al. The flat values were five times too small at prime ages and the wrong shape
AE A15	χ_H against Abramson and Van Nieuwerburgh	Their \$2,500 per household month for the literally homeless gives 0.4286. Note the direction: R2 and R3 called 0.252 too high and the best structural estimate is higher still
AE A16	Homelessness definition	Literal homelessness, and the target is now on a household basis, 0.399%
R1 4(c)	Administrative cost of drug testing	In the government budget. Detection requires a test, so effective detection is test × detect
R1 1(b), AE A10	Treatment take up	Endogenous, at a utility cost, with a target. NSDUH validates the *structure*: the three largest barriers are beliefs and preferences, not prices
R2 D1	Housing grid	Identified from the ACS B25042 size distribution rather than from housing cost burdens, which become untargeted validation

Ref	Item	What closed it
,	b_1, the SUD Stone–Geary shifter	Given a target, the Milby abstinence rate among the homeless with SUD. No longer asserted
,	g, the treatment effect	Closed enough. One clean randomised estimate, an odds ratio of 1.22 for contingency management. Stays a reported range rather than a point
,	SUD target provenance	3.68% is moderate or severe *drug* use disorder at ages 26+, reproduced exactly from the NSDUH microdata. Needs stating in the paper, since NSDUH's headline of 16.6% includes alcohol
R1 2(c)	unemp | SUD	**Measured at 16.25%**, replacing an asserted 26.9% whose stated source could not have produced it. The channel was 2.25 times too strong

Ref	Item	Plan
R2 C3, AE A6	Parental anxiety; decorative intergenerational language	**Decided: the warm glow.** At one parental age the value gains $\nu\,\mathbb{E}_{x'\|x}[V_1]$, using the $\rho_x = 0.5$ process §4.1.3 already specifies but never uses. The parent cannot influence the draw, so it is additive and moves no decision: it shifts welfare levels and hence the policy comparison, which is what R2 predicts. ν cannot be calibrated, since it moves no behaviour, so it is a robustness range. Scaffolded inert at $\nu = 0$
R1 3(d)	Transition paths	**Solver written and validated end to end.** Damping is now adaptive, which lifted the contraction from 0.82 to 0.51 per iteration. What remains is a production run, and its cost
R1 2(e)	Replicability of the moments	An appendix defining each moment's model counterpart. Mechanical writing that nobody has started

Ref	Item	Plan
R1 sm. 1	Maximum experimentation age	A diagnostic run. If dropping it piles use onto the old, that has to be explained rather than assumed away

Ref	Item	Why it is still open
,	**The calibration has not converged**	The real blocker. It gates the ladder, the externality decomposition and the whole second stage
R2 C2	ω_x, the composition weight	Searched for and not found. The literature supports a difference in *sign* by facility type, not a magnitude. Should be presented as a sign restriction with a range
,	$\kappa_{\phi m}$ level	Constant in age, which is settled, but the level is a judgement across three populations measuring different things. Sensitivity at 3.2

So: ten items closed, four decided and waiting on execution, and four genuinely open, of which only one, the calibration, is on the critical path.

9.8 The critical path

Three things gate almost everything above.

1. **The first stage calibration converging.** Unblocks R1 2(a), 2(b), 2(c), 1(d), R2 C4, R3 M2b, AE A12 at once.
2. **The second stage calibration**, which is built and needs only a converged benchmark to run against. This is R2 C1, R3 and AE A13, what all three independently call the crux, and it is a partial equilibrium cohort experiment, so it does *not* wait on the transition solver.
3. **The counterfactuals.** Unblock AE A9, A18, R2 C6, R3 m3, R1 3(c), 4(d). Then the transition-path solver for R1 3(d), which is the one large build left and affects only the aggregate policy experiments.

Everything else on the outstanding lists is either writing or a single cheap run.